

Reciprocal relationships, reverse causality, and temporal ordering: Testing theories with cross-lagged panel models

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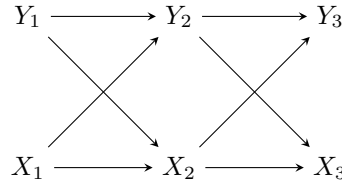


Figure 1: Directed acyclic graph representation of classic CLPM

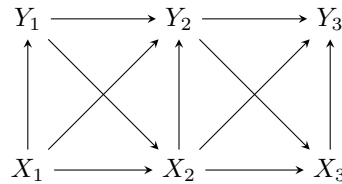


Figure 2: CLPM with contemporaneous effects

CHUCK

Imaginary data

- Reality and theory move at one recursive (really continuous) pace, but the practical limitations of empirical data collection result in snapshots of data—often captured at much lower resolution than our theory.
- For the purpose of illustration, let us assume that in both cases we have three waves of panel data with yearly waves of observation.
- The empirical observation periods clearly do not align with the theoretical causal process.

- Theory can imply whether you can plausibly isolate or rule out reciprocal effects or reverse causality.
- For our first example of the reciprocal effects of fear of crime and disorder perceptions, we expect that both processes of updating occur more rapidly than over course of a year—more quickly, even, than could be captured with *daily* data. Convincingly ruling out reciprocity would require biophysical measurements in a laboratory, an approach which would come at a cost of construct and external validity (i.e., real disorder out in the world).
- For our second example of reverse causality between employment and offending, we expect the process to occur less quickly but there is still plenty of time for feedback in yearly observations. If we decreased gaps between waves—perhaps to daily observations—we could convincingly rule out reverse causality.
- More generally, we can imagine the discordance between our theoretical model and our empirical data collection as a missing data or latent variable problem as illustrated in FIGX.
- As noted, in the real world, time is continuous, so one might imagine there exists an infinite number of these “missing waves” between observations.
- One must always make some assumption about the way variables and relationships operate in these unobserved intervals.

I need to think harder about this “missing data” thing about whether it alone really introduces the contemporaneous reciprocity thing or if this really only occurs in the context of demeaned variables / fixed effects.

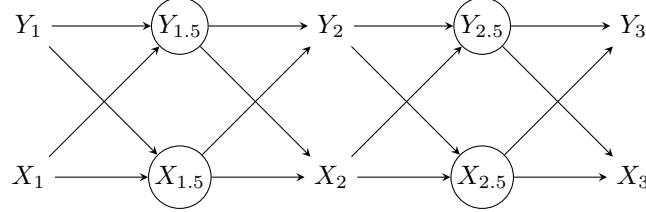


Figure 3: A theoretical model of cross-lagged effects with incompletely observed periods.

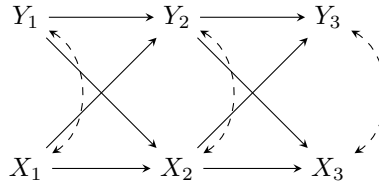


Figure 4: A theoretical model of cross-lagged effects with ambiguous contemporaneous relations.

SEM diagrams

Data observed, and thus you must estimate, from discrete often arbitrarily spaced waves; this produces reverse causality

Once the theoretical model has been established, one can proceed to developing a statistical model that accounts for the constraints from the research design. Here, path diagrams from the structural equation modeling (SEM) tradition become useful.

- SEM diagrams encode both structural relations between variables indicated by single-headed arrows pointing from cause to effect and unanalyzed, often “nuisance” associations, indicated by double-headed arrows pointing between two variables.
- Our recommendation is that the structural part of the SEM diagram should be recursive, corresponding as closely as possible to the theoretical model. Reciprocal simultaneous effects should generally be avoided as they require strong identifying assumptions, e.g., valid and relevant instrumental variables (Green 2003; XXXX).
- When the timing of observation results in ambiguity regarding the causal direction within any given time period—such as when periods $t = 1.5$ and $t = 2.5$ were not observed in Fig XXX—we can encode this ambiguity as an unanalyzed covariance.
 - Work out the diagrams; see if they are the same? Probably are!

FIGURE: Time estimator; the classic CLPM

- Explain the classic CLPM as an estimator

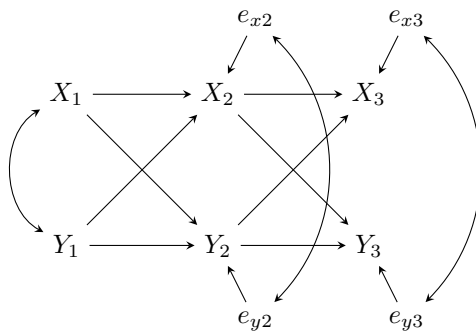


Figure 5: SEM path diagram representation of three-wave CLPM

Thiago: Bollen & Pearl 2009 provide a nice discussion about the notational relationship between SEM path diagrams and DAGs. We should also look at Imai & Kim’s three papers on causal inference with panel data. They use DAGs and they’re so well written. I don’t think they discuss reciprocal effects at all (could be wrong), but we could rely on them from a notational point of view in using DAGs to depict temporal processes.

How it can go wrong!

- One para introducing the three key issues:
 - Temporal order and timing
 - Unobserved heterogeneity
 - Lack of inter-temporal variation